

ORION – Space Scene Generation Framework



ORION Space Scene Generation framework is a system that covers all space scene generation aspects, from procedural planets & spaceships to any relevant special effects, with support for all pipelines.

The ORION asset can be upgraded from Sky Master ULTIMATE at a lower price, offering the possibility to combine its systems with **Sky Master ULTIMATE** for [full volumetric planetary clouds](#) and atmospheric effects, that are best suited for flying through the cloud volume.

This is the first Beta phase release of the system, a lot more aspects and features are going to be added in the future releases, as well as conversion of some of the image effect based modules (e.g. volumetric atmosphere) to the URP and HDRP pipelines.

For help with using the system and for using it with [Sky Master ULTIMATE](#) Weather suit, please visit **ARTnGAME** [Discord channel](#) and the [ORION forum thread](#). For reference on using the system may use the [ORION Tutorial videos on Youtube](#).

ORION features (TBD are to be decided if will eventually be included in the system):

- **Procedural planet generation** (All pipelines)
- **Lens flares** from multiple brightness sources (SRP - **TBD** URP-HDRP)
- **Black hole** system (All pipelines) - **Galaxy** formation generator (All pipelines)
- **Volumetric space dust** emulation (SRP - **TBD** URP-HDRP)
- **Dynamic Background Nebula** (All pipelines)
- **Planet atmosphere** with light scattering (SRP - **TBD** URP-HDRP)
- **LOD based planet generation** and traversal in run time (All pipelines)
- **Procedural Spaceship** generator (All pipelines)
- **Dynamic planetary cloud** systems (All pipelines)
- **Dynamic procedural planet** shaping and motion (All pipelines)
- **Combination with Sky Master ULTIMATE** for full volumetric planetary clouds (All pipelines)
- **Lightning** system for planets with multidirectional lighting (All pipelines)
- Fluid based **dynamic planet texture generation (Ideal for Gas Giants)** (All pipelines)

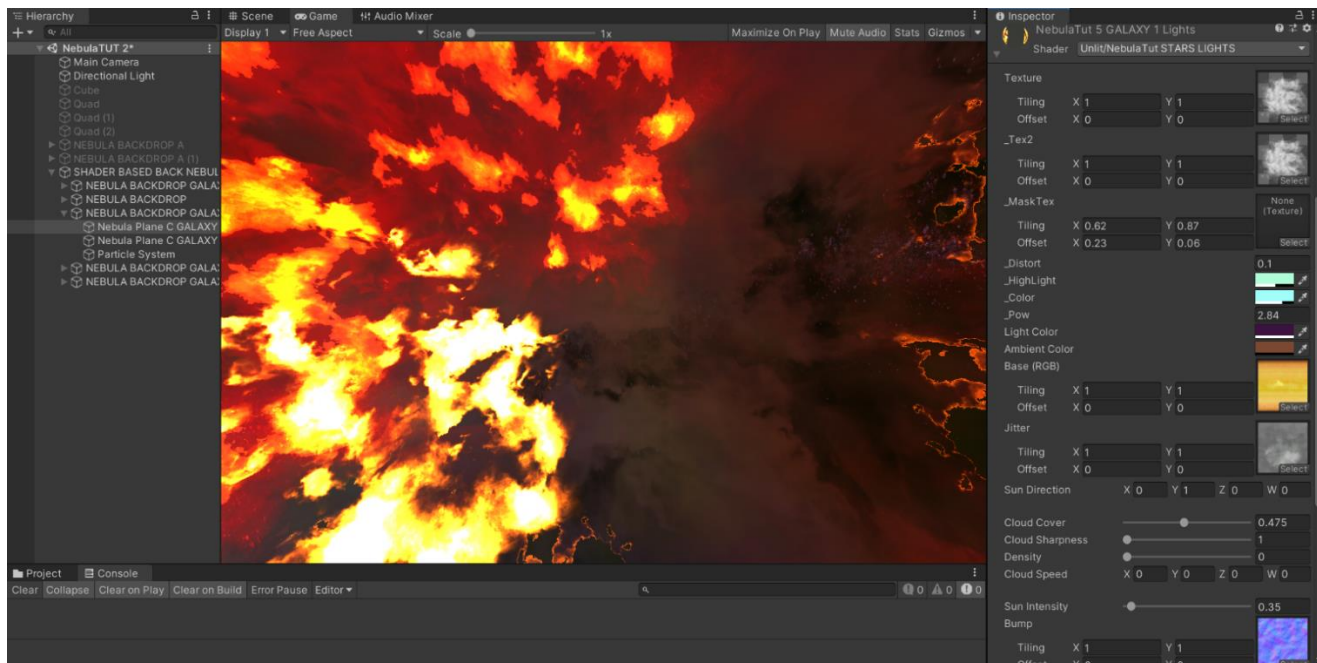
In the following the various ORION system will be presented with information on use and tips based on pipeline used.

Nebula – Shader Based Background version



This module is based on a shader group that allow the superposition of effects on the inside of a sphere that **can be placed around the map as background**. The sphere must be scaled to reach outside the play area and this effect is not suitable for fly through the fluid and gas volume.

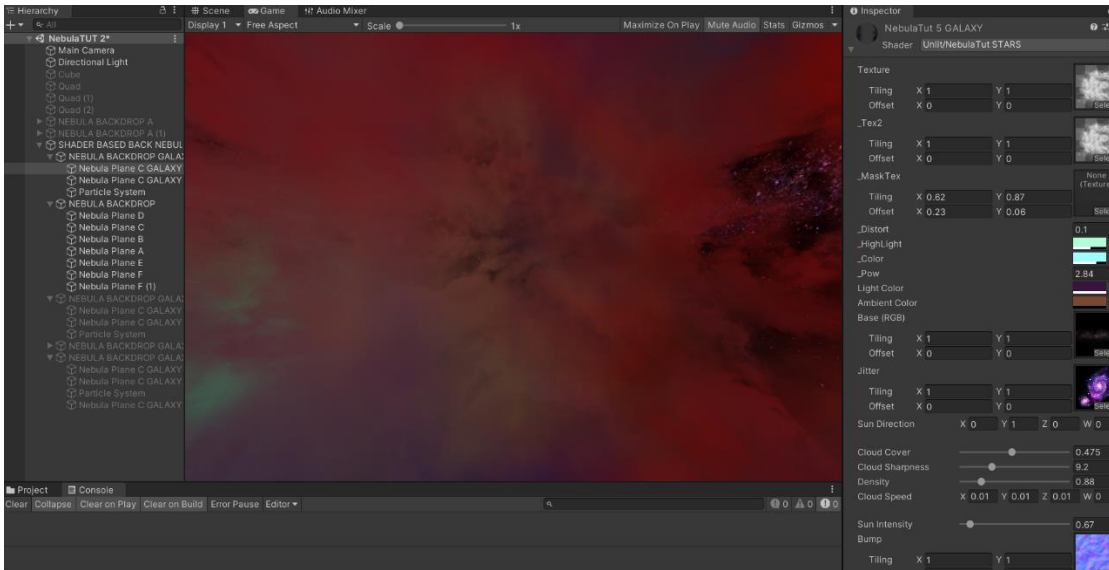
Below are presented the shader and options, the “STARS Lights” version includes a galaxy view behind the gas clouds, as well as possibility to affect the edges of the cloud with extra lighting effects, the galaxy texture (or any other backdrops to appear behind the clouds) must go in the “Base” map slot. The “Density” scales the galaxy map and “Cloud cover” and sharpness regulate the amount of galaxy influence to the gas clouds.



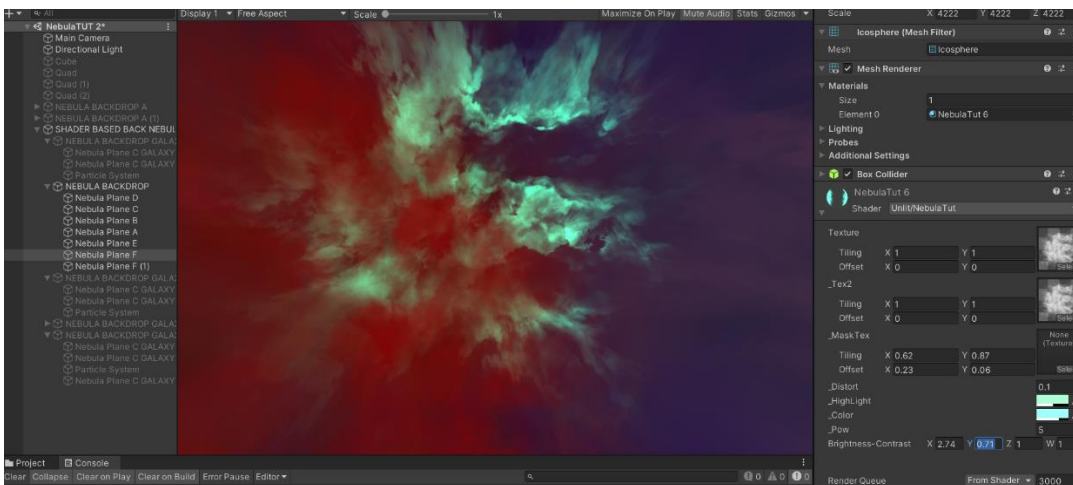
With “Increase galaxy effect” can further regulate the galaxy intensity in the clouds. The “Increase Cloud Speed” affects the cloud motion velocity, use zero to stop the cloud motion. The Edge controls affect how much the edges of the cloud – galaxy are colorized and the distance of the colorization.

Increase Galaxy Effect	X 1	Y 1	Z 1	W 2.9
Increase cloud Speed	X 155	Y 0	Z 0	W 0
Brightness-Contrast	X 1	Y 1	Z 1	W 1
edge Controls A	X 2	Y 1	Z 0.54	W 1
edge Controls B	X 15.2	Y 0.05	Z 0.53	W 0
edge mode Controls	X 0	Y 0	Z 0	W 0
Render Queue	From Shader 3000			

The “STARS” version of the shader presented below includes only the cloud and galaxy, without edges control and can be used as a faster option if no edge control is needed.



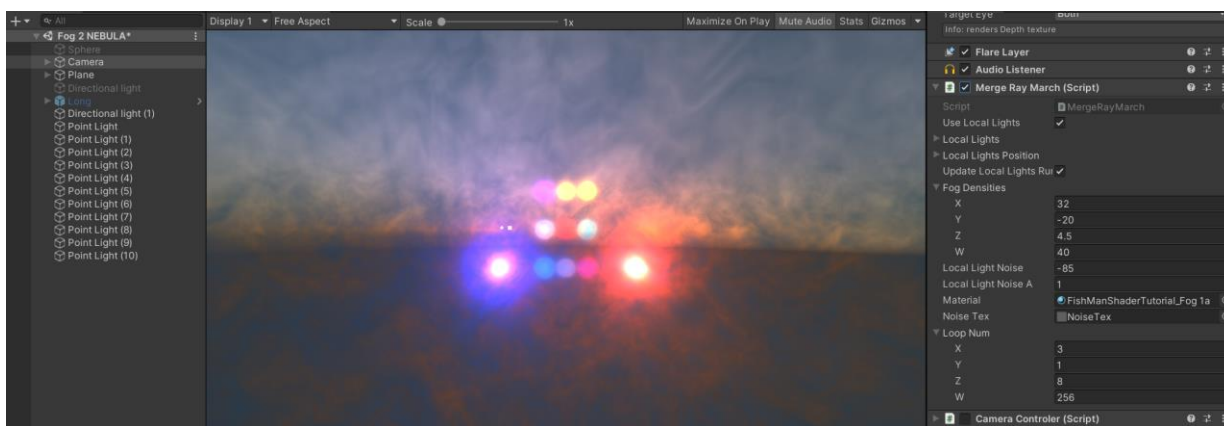
The “NebulaTut” version of the shader presented below includes only the cloud, without the galaxy or edges control and can be used as a faster option if no galaxy and edge control is needed, also is great for combining with the above two versions for extra effect and cloud colors, e.g. the green extra cloud shown in the following image.



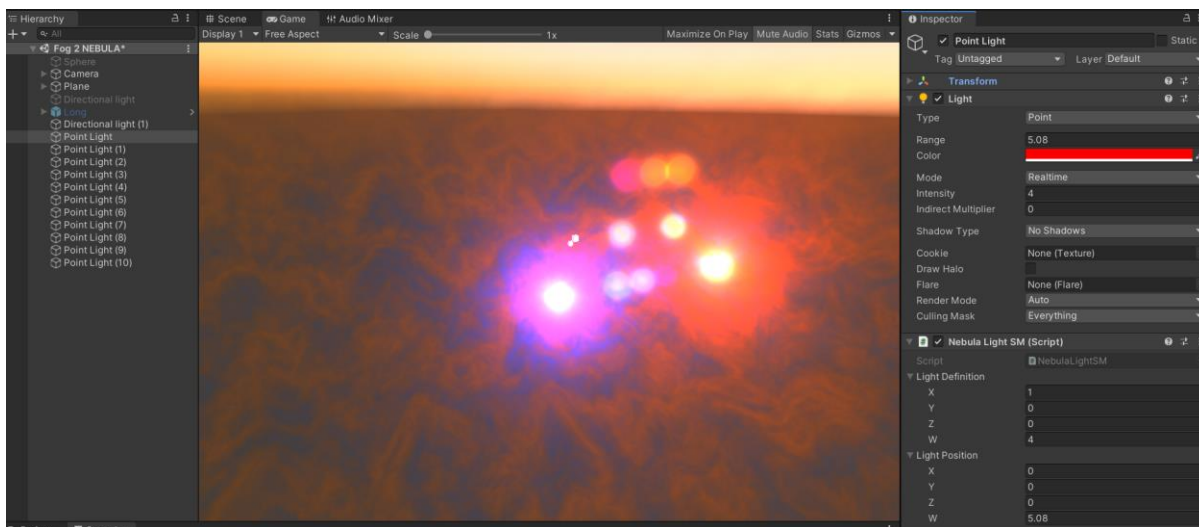
Nebula – Image Effect Based fly through version (Standard Pipeline, HDRP-URP under development)



This is a fully volumetric nebula system, works currently in Standard Pipeline and eventually will be converted for URP-HDRP. The system is setup by the “Merge Ray March” script on the camera, configured with the fog material in the “Material” slot. The script is a general volumetrics solution and can be configured with various other materials (e.g. there is also a Stars volume available using the relevant material)

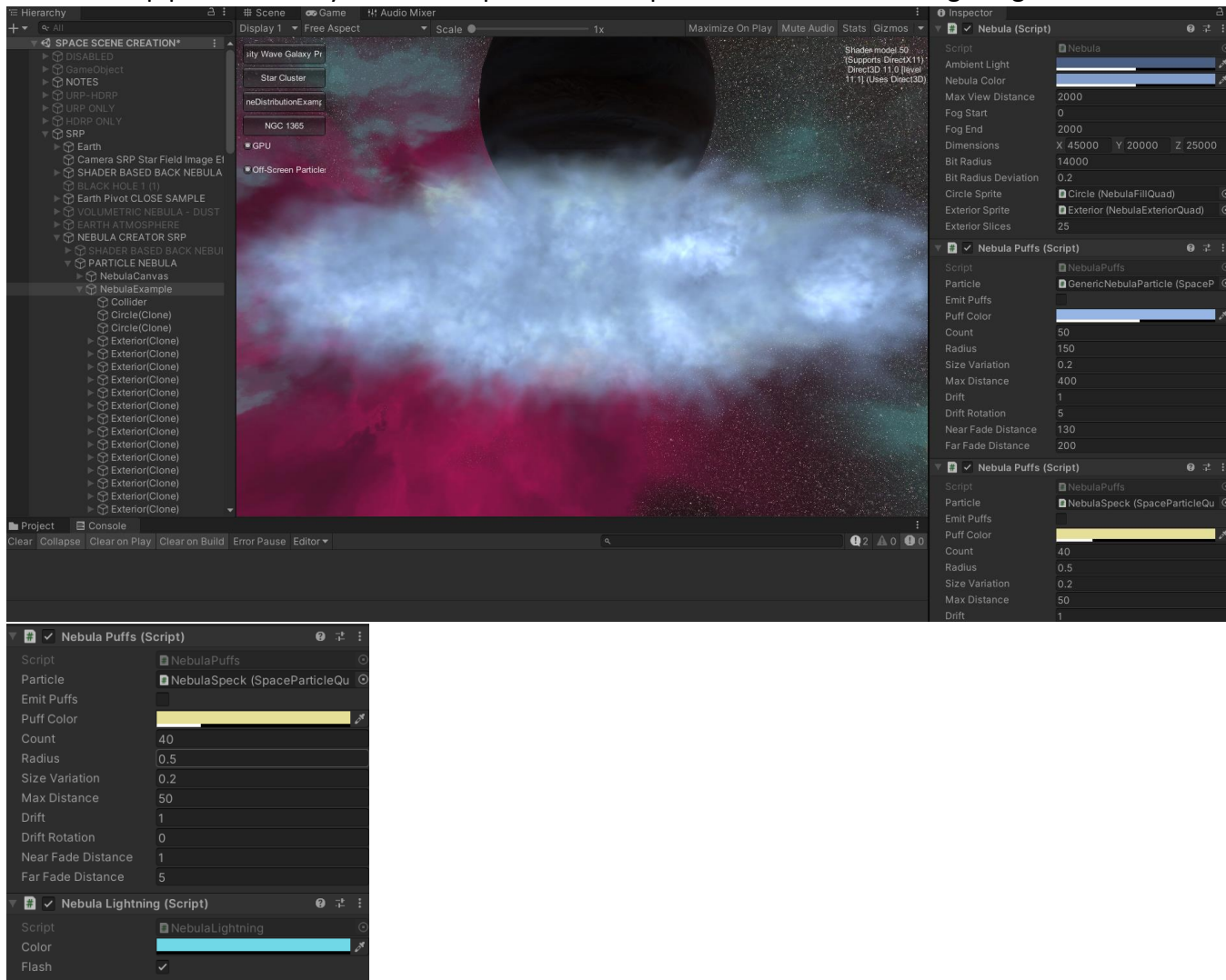


The fog look and density can be regulated by the densities and noise variables. The system can also render volumes of local point lights, to enable this for a point light, add the “NebulaLightSM” script on the light as shown in the image below.

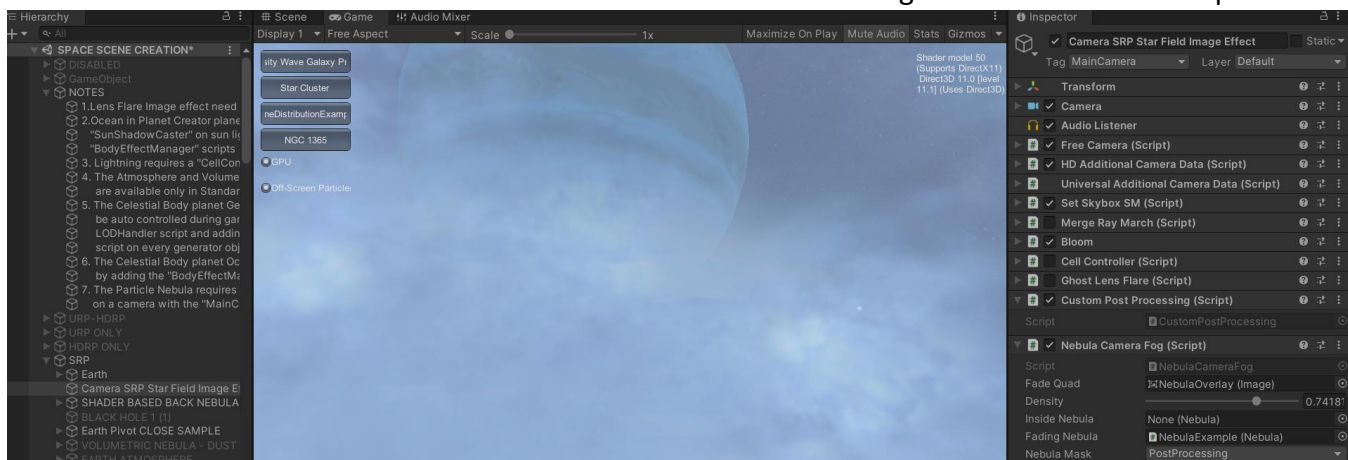


Particle Based Nebula

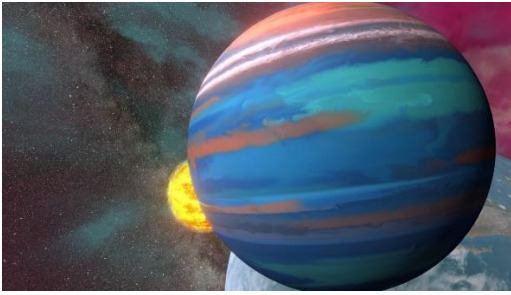
This module is a particle based nebula that can also support fly through the volume emulation, this system works in all pipelines. The system is setup with the scripts shown in the following image.



The system also requires the “NebulaCameraFog” on the camera, which should also be tagged as “MainCamera”. This is used so the system can transition from the empty space to the inside the nebula space. The Nebula shown above must be referenced in the Inside and Fading Nebula slots in the script.

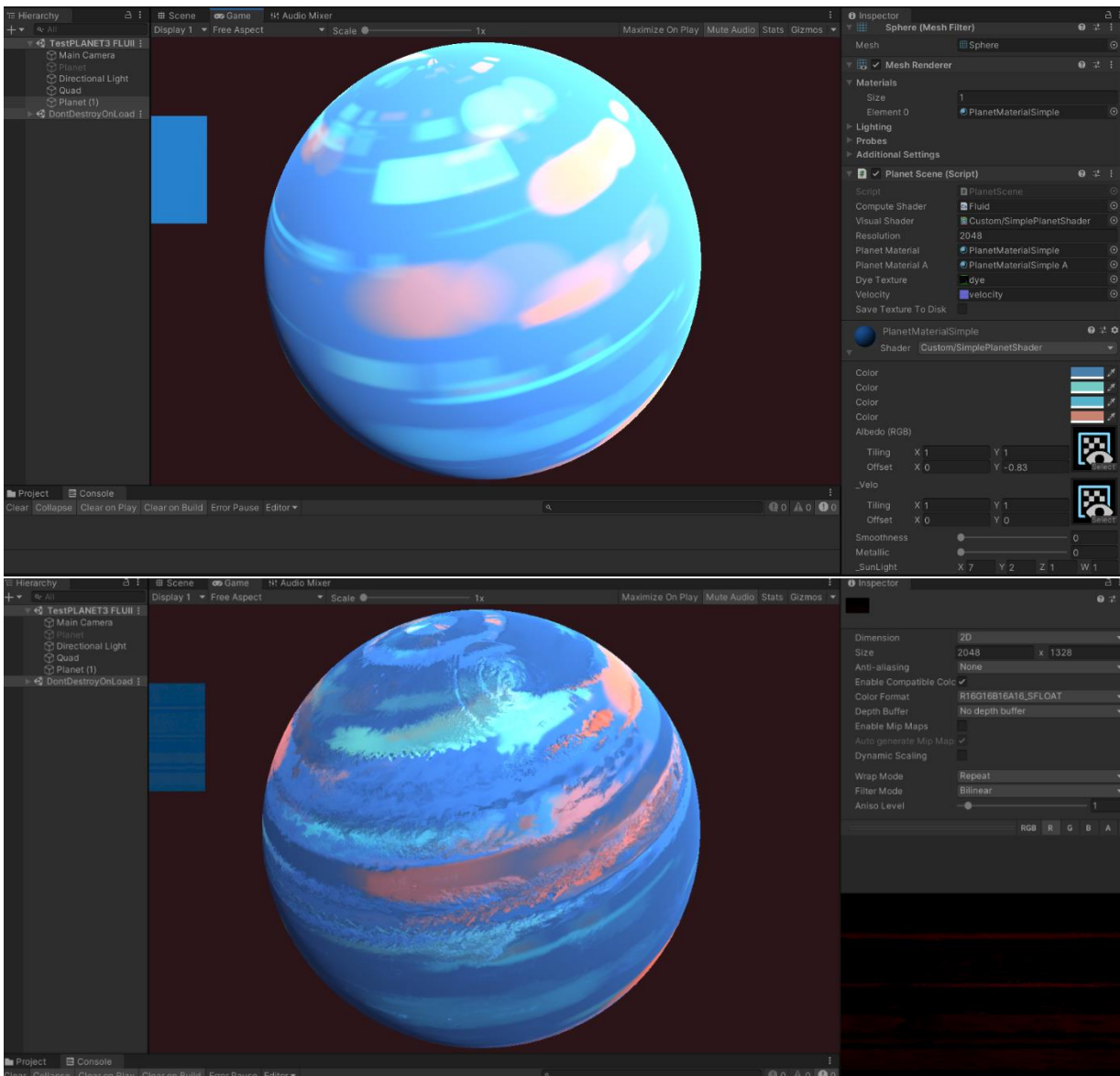


Gas Giant Planet Generation with Fluid based simulation

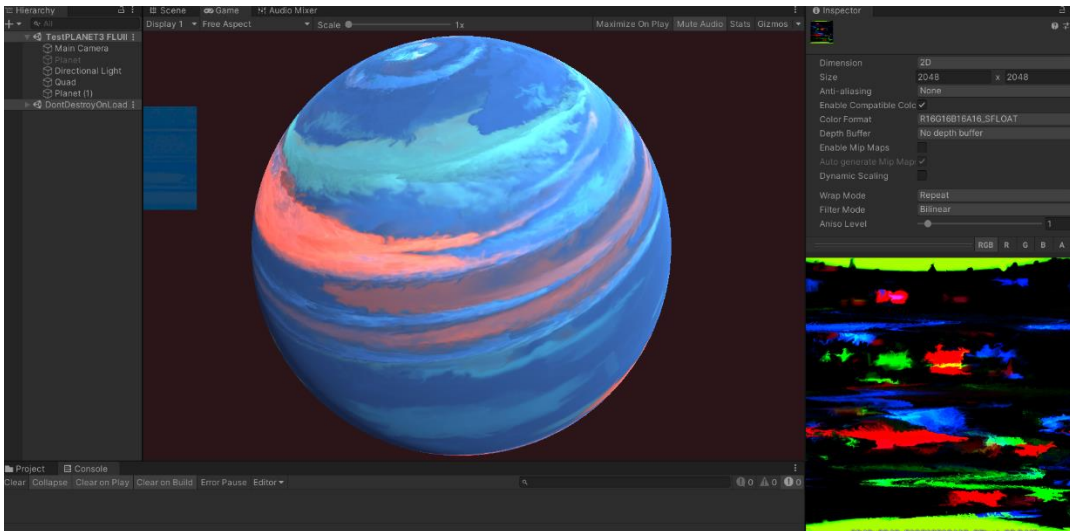


This module can be used to create and store a planet texture created using fluid mechanics. This is ideal for creating gas giant like planet textures. To use this system place the “Planet Scene” on a sphere object and assign the material with the “Simple Planet Shader” to the sphere and the “Planet Material” slot in the script.

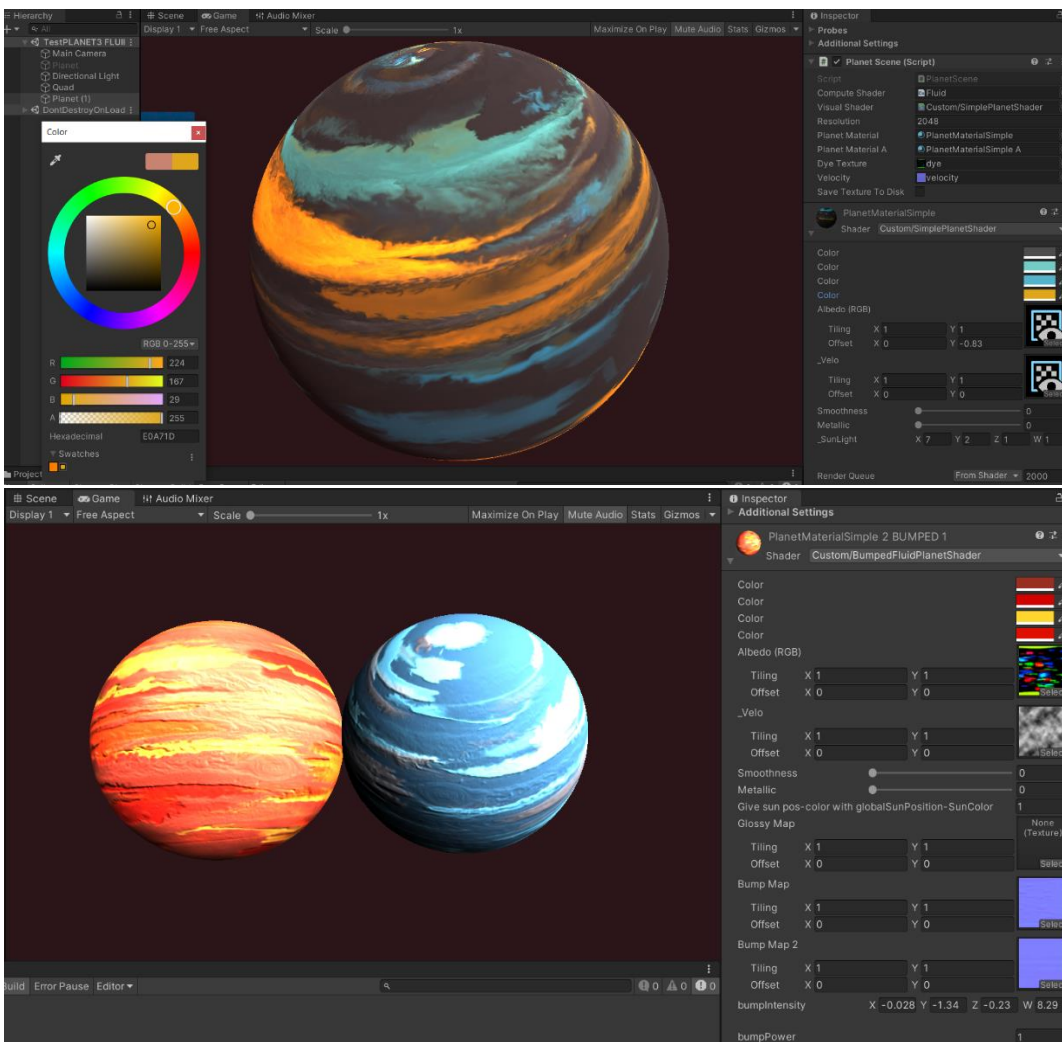
Then enter play mode to enable the fluid simulation and press “O”+”L” keyboard buttons to apply a dye to the planet surface and “O”+”P” buttons to do the fluid simulation. Then can press only the “O” button to smooth out the effect.



Then use the “Save Texture to Disk” checkbox to save the texture in assets. The texture saved is shown below.



This texture can then be used with the “SimplePlanetShader” shader in a material to present the planet in the game. A more advanced shader called “BumpedFluidShader” is also available, for adding extra bump mapping on the planet surface.

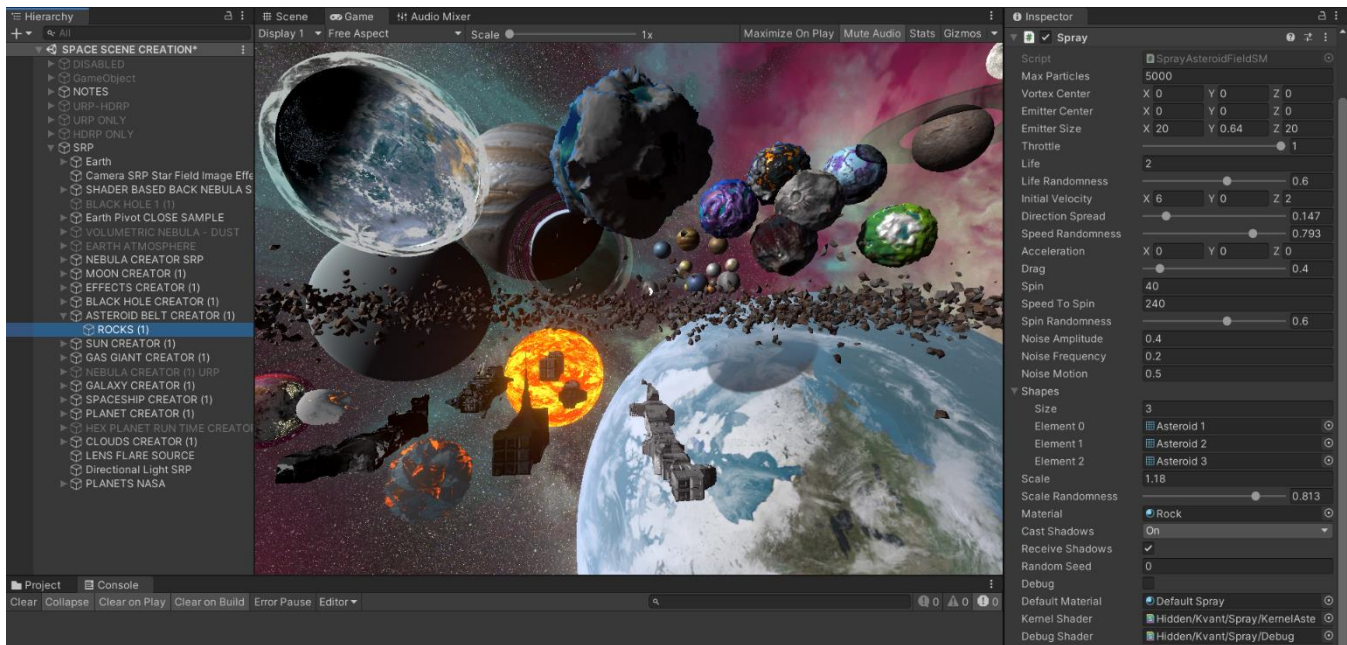


NOTE 1: The system currently works only in Gamma color mode, thus is possible to create the textures in Gamma mode and use with the planet shader in Gamma or Linear Modes afterwards.

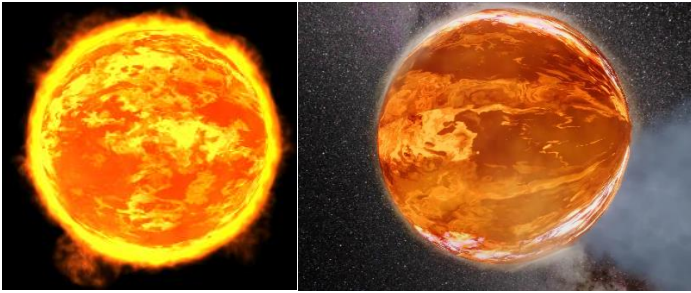
Asteroid fields



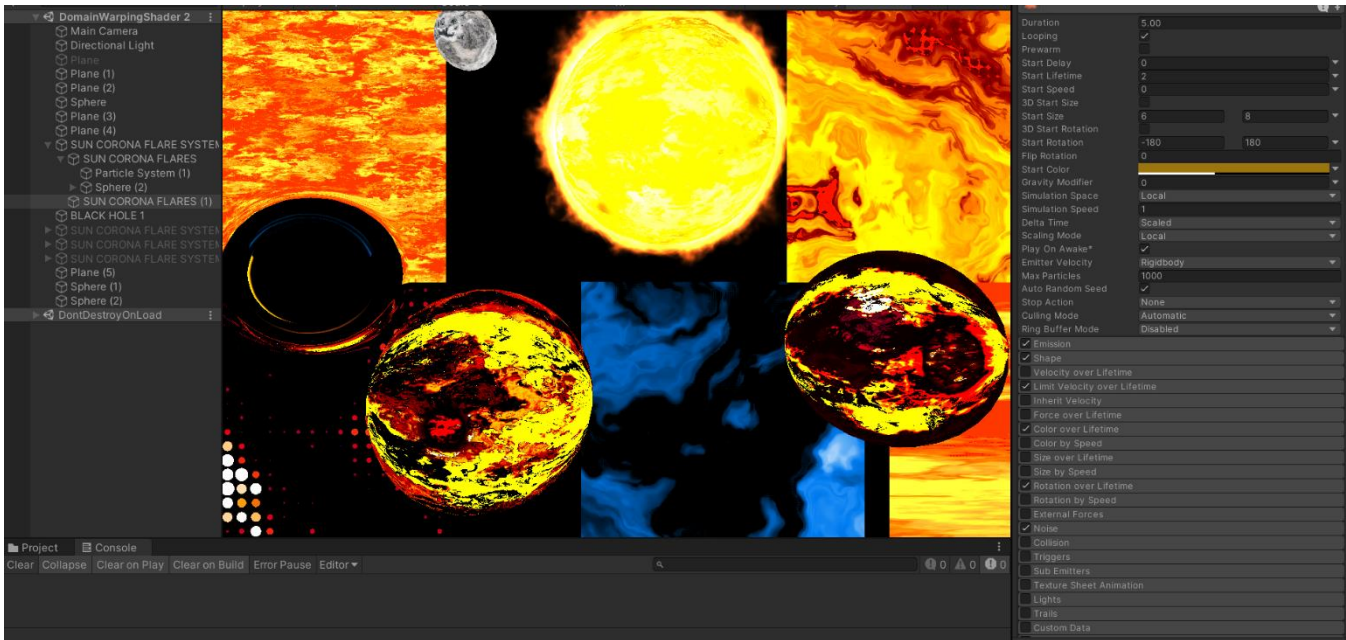
This module is used to create round formations of mesh particles for emulation of asteroid fields. Note that there is no physics or collisions involved in the system. The system is setup using the “Spray” script and a mesh in the “Shapes” list and a material in the Material slot. The material must be changed based on the pipeline.



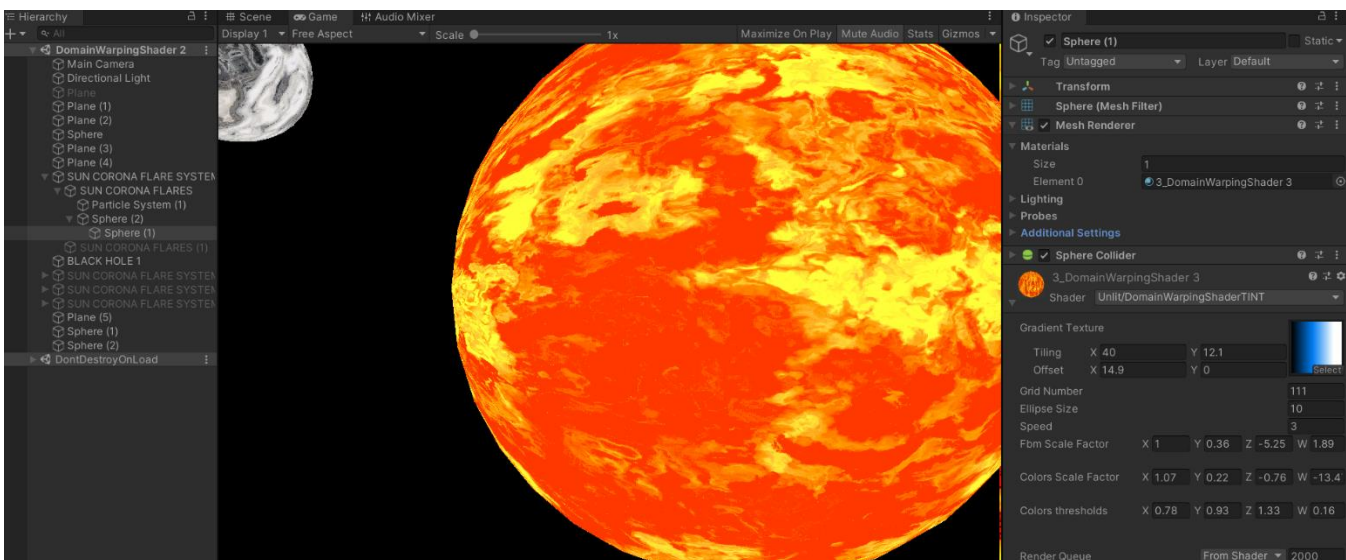
Dynamic Sun Simulation



This module consists of a shader that emulates fluid motion, as shown in the textures below and a particle system that creates the sun flares.



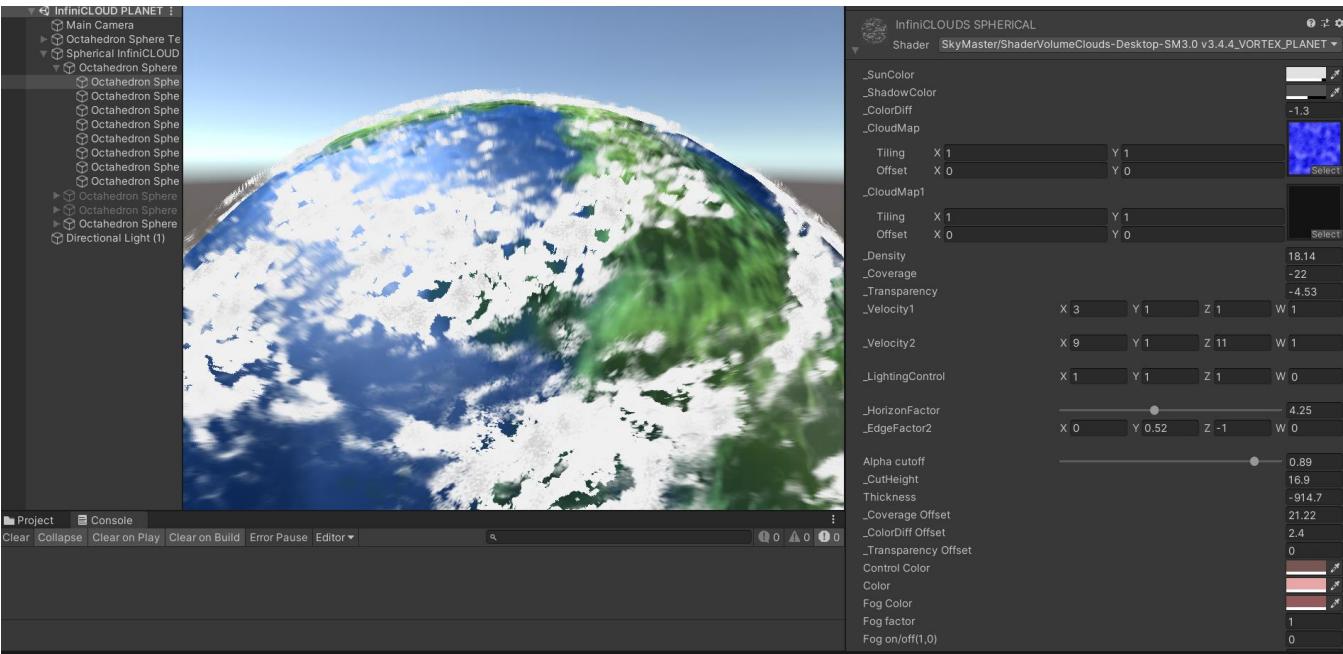
The fluid shader can be directly regulated on the sun material, as shown in the following image.



Volumetric Shader Based Dynamic Clouds



This module creates a volumetric looking cloud bed around a planet, using stacked spheres with the InfiniCLOUD shader applied, the cloud shader is shown un the following image. The main setup is by the two Cloud Map textures, one is the cloud noise. Then regulate directly the density and coverage for getting various coverage and the ColorDif variable for regulating the lighting.



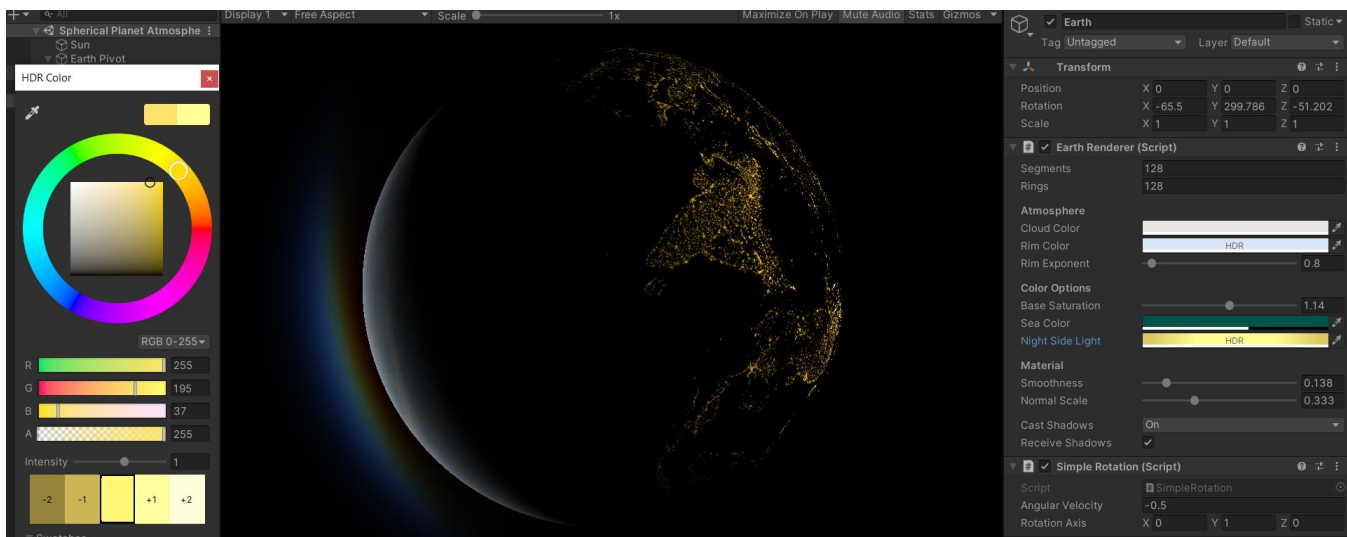
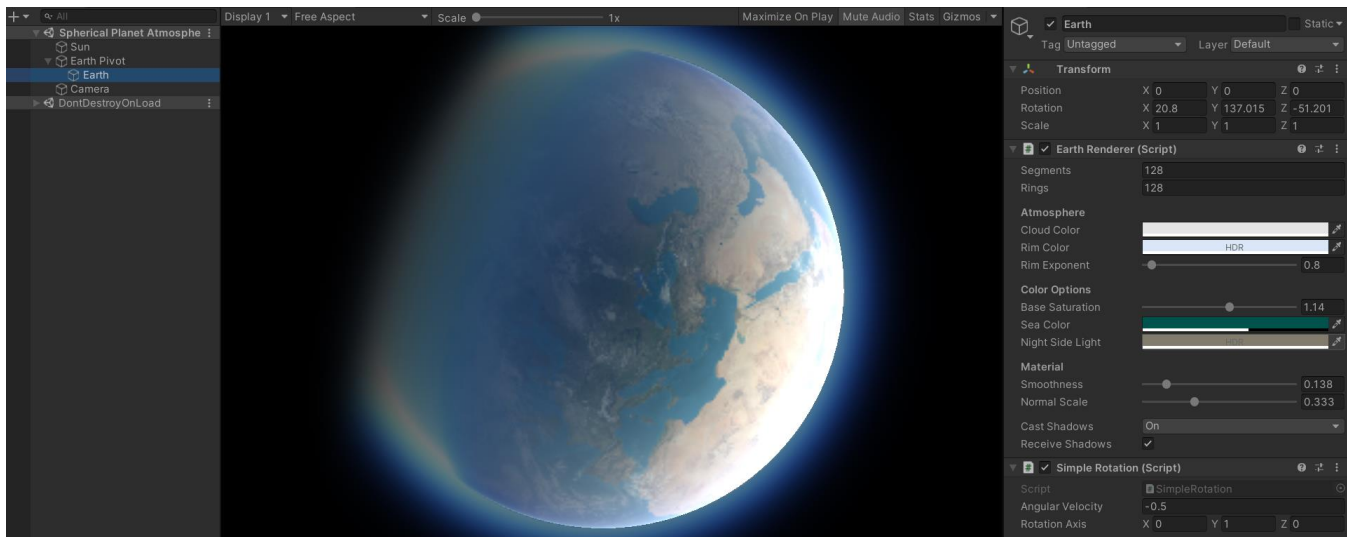
The clouds can also be controlled by the PaintMap map transparency, for further variety in the cloud density based on this texture. Use the tiling and offset to scale and place the texture effect on the cloud bed.



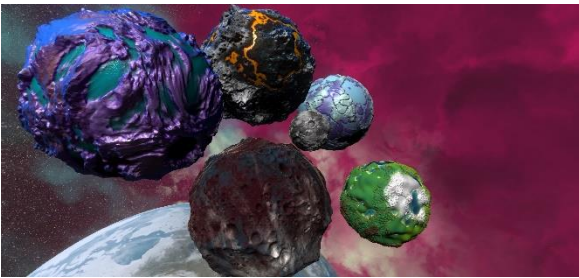
Earth Day – Night Shading



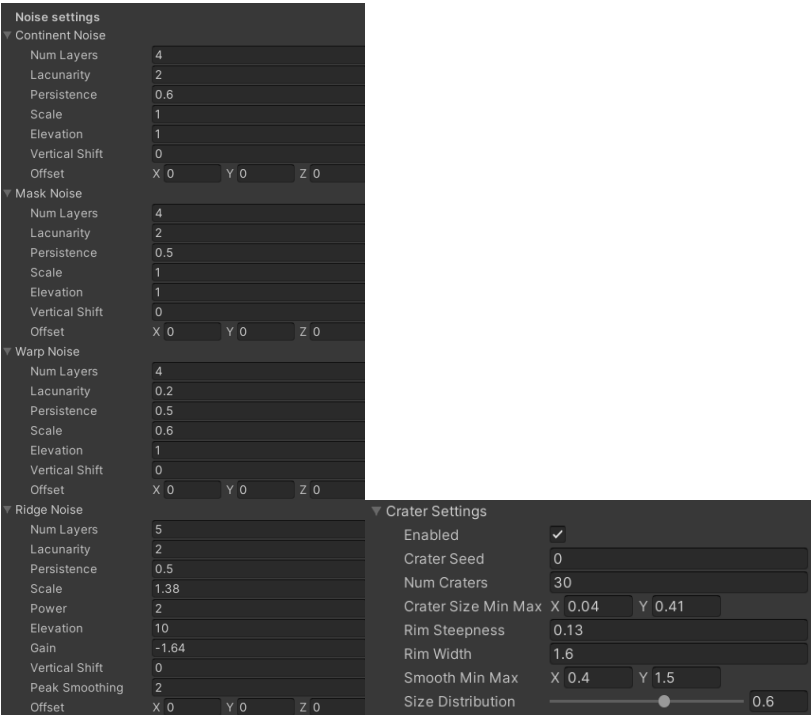
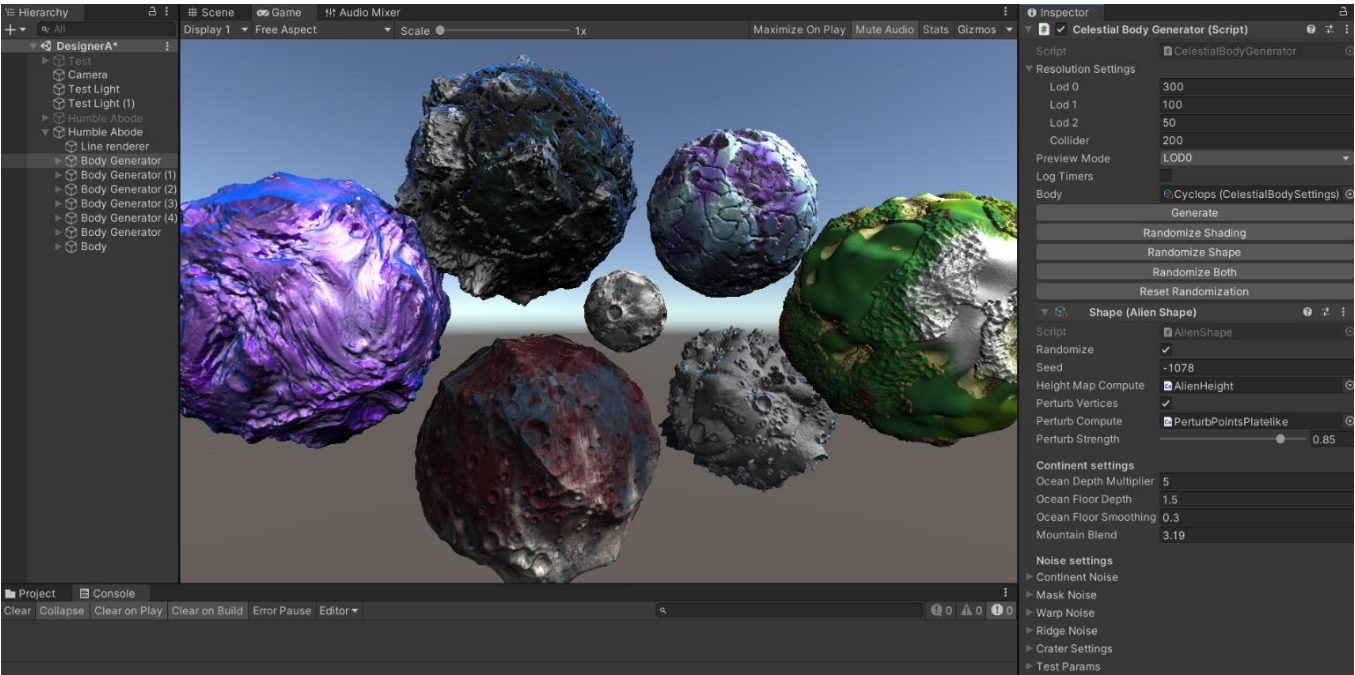
The system includes a shader that emulates night time lights on the planet, integrated in the “EarthRenderer script”. The script will create the earth dome and can adjust the earth lights by the “Night side Light” variable.



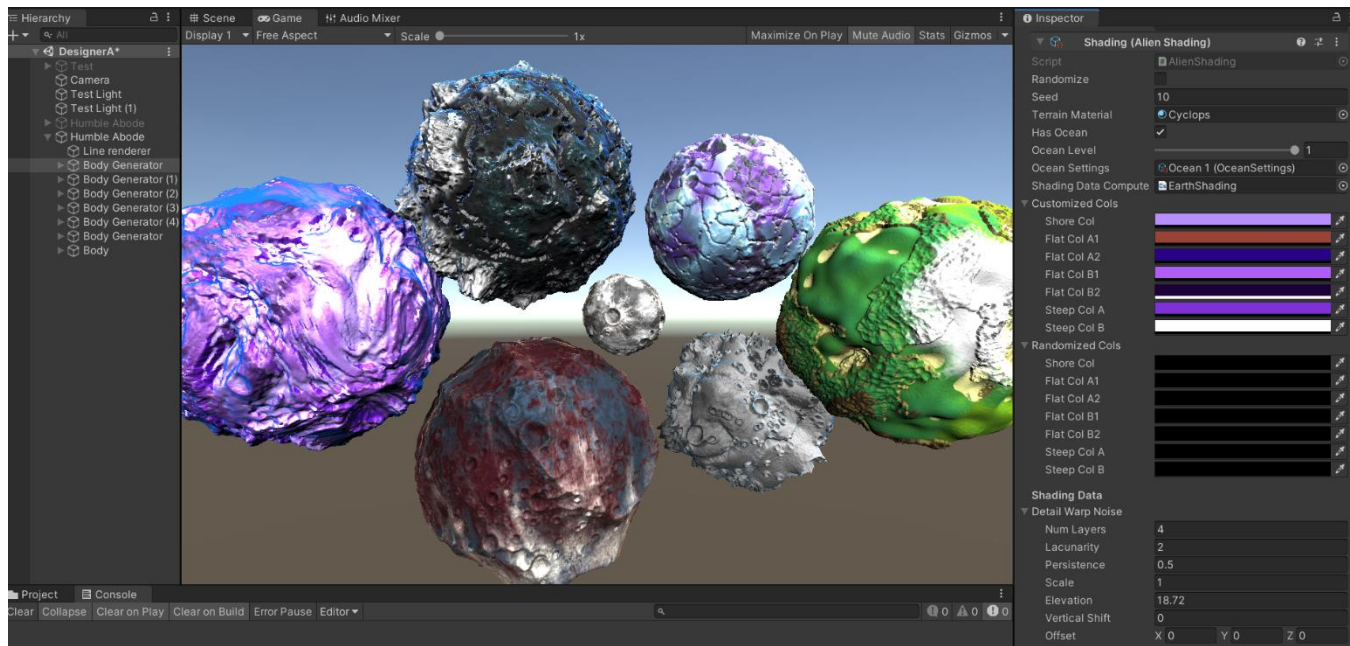
Planets Generator



This module can be used to generate procedurally planet surfaces, using various noise, shaping and coloration options. The main settings for each planet are held in the “Body” slot and should be changed for each new planet. The various noise options are shown in the following images.

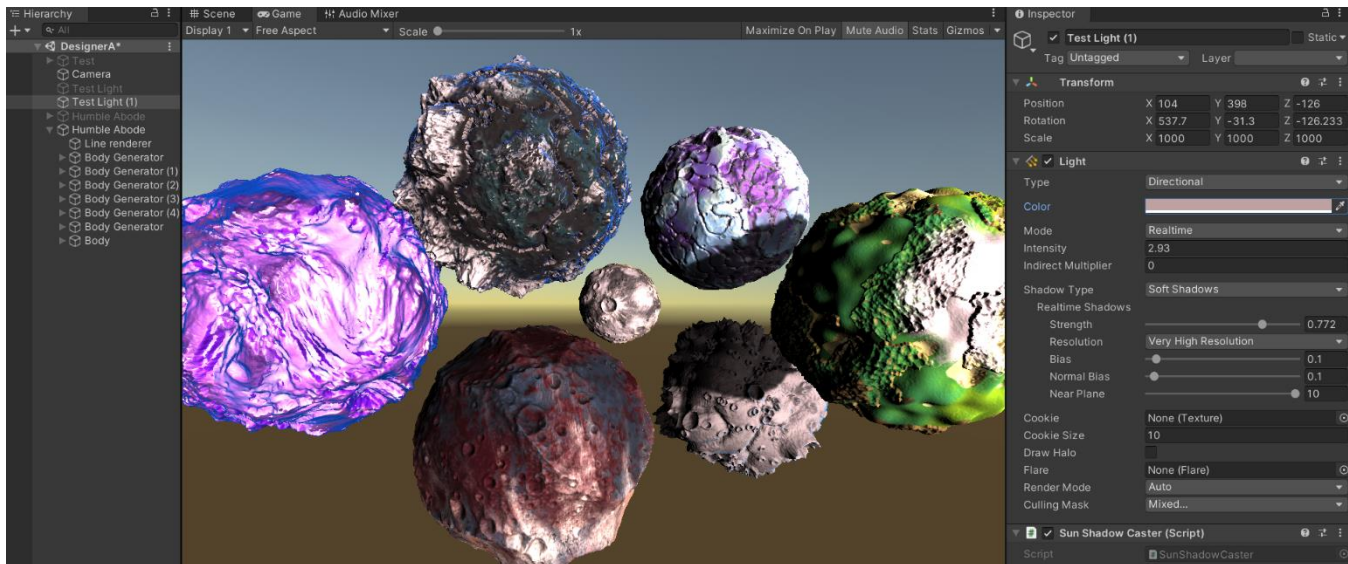


The various coloration options are shown in the following images, the oceans material can also be configured in the “Ocean Settings” slot. The main planet material is set in the “Terrain Material” slot.

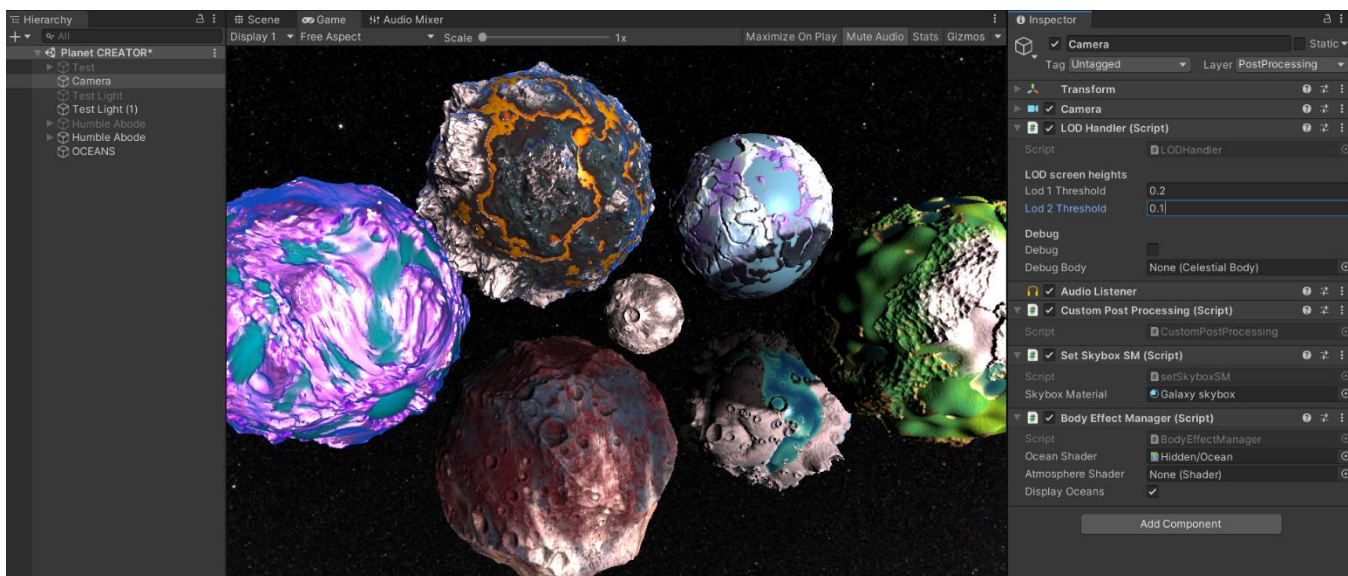


▼ Detail Noise	
Num Layers	4
Lacunarity	2
Persistence	0.5
Scale	0.89
Elevation	-1.25
Vertical Shift	0
Offset	X 1.35 Y 0 Z 0
▼ Large Noise	
Num Layers	4
Lacunarity	2
Persistence	0.26
Scale	0.425
Elevation	1
Vertical Shift	0
Offset	X 0 Y 0 Z 0
▼ Small Noise	
Num Layers	4
Lacunarity	2
Persistence	0.5
Scale	5
Elevation	1
Vertical Shift	0
Offset	X 0 Y 0 Z 0
▼ Noise 2	
Num Layers	4
Lacunarity	2
Persistence	0.5
Scale	0.6
Elevation	1
Vertical Shift	0
Offset	X 0 Y 0 Z 0
▼ Warp 2	
Num Layers	4
Lacunarity	2
Persistence	0.5
Scale	1
Elevation	5.37

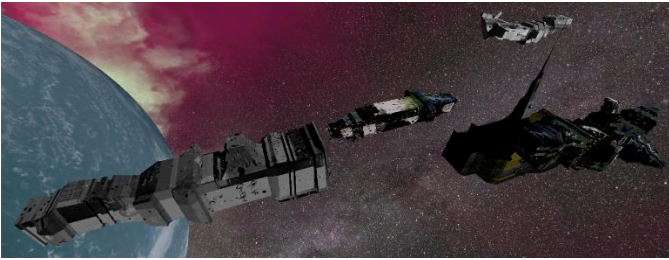
A number of extra scripts may be used to further regulate the system. Use the “Sun shadow caster” on the scene directional light to allow shadowing of the planets.



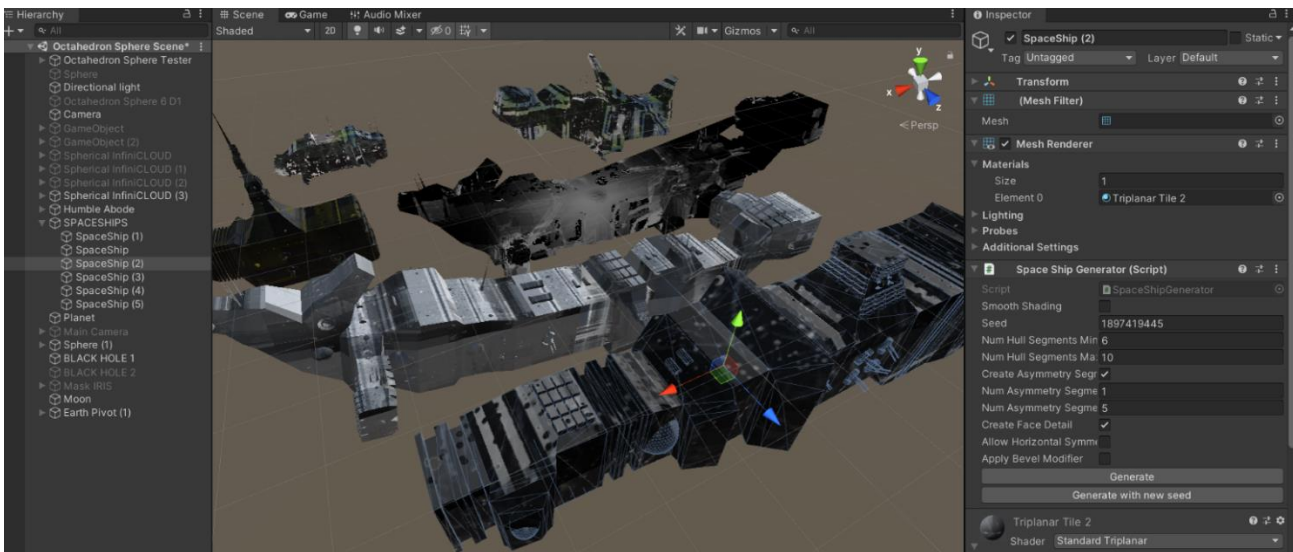
Use the “Body Effect Manager” script to apply the oceans shading as showing the image below, the oceans are rendered directly with using a mesh in the scene. Use the “LOD Handler” script to regulate the LODs activation at run time for optimization. The thresholds are the portion of the screen the planet covers beyond which the LOD is activated. For example if the planet takes up less than 20% of the screen, LOD1 will be enabled and when take up only 10% LOD2 will be activated in the sample settings in the following image.



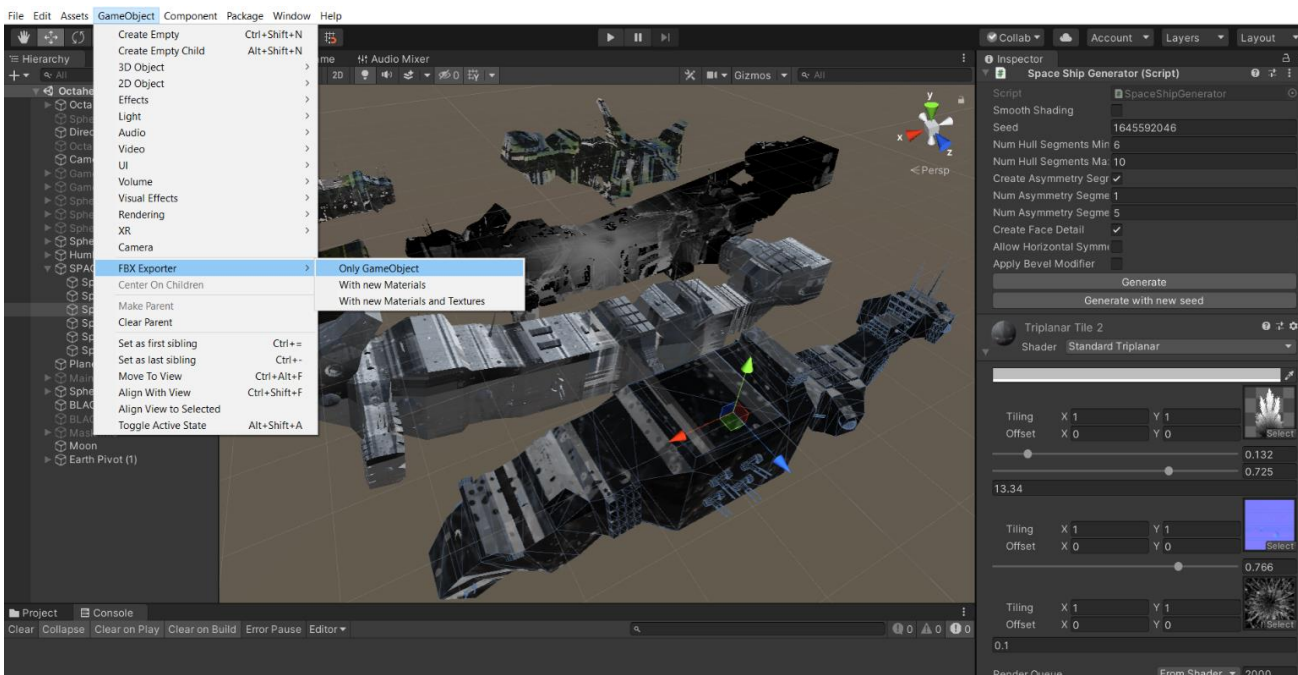
Spaceship Generator



This module can be used to generate procedurally an infinite number of spaceships. Each spaceship can be created using the “SpaceShipGenerator” script. The system does not create UVs, thus a triplanar shader is used to apply material to the spaceships.

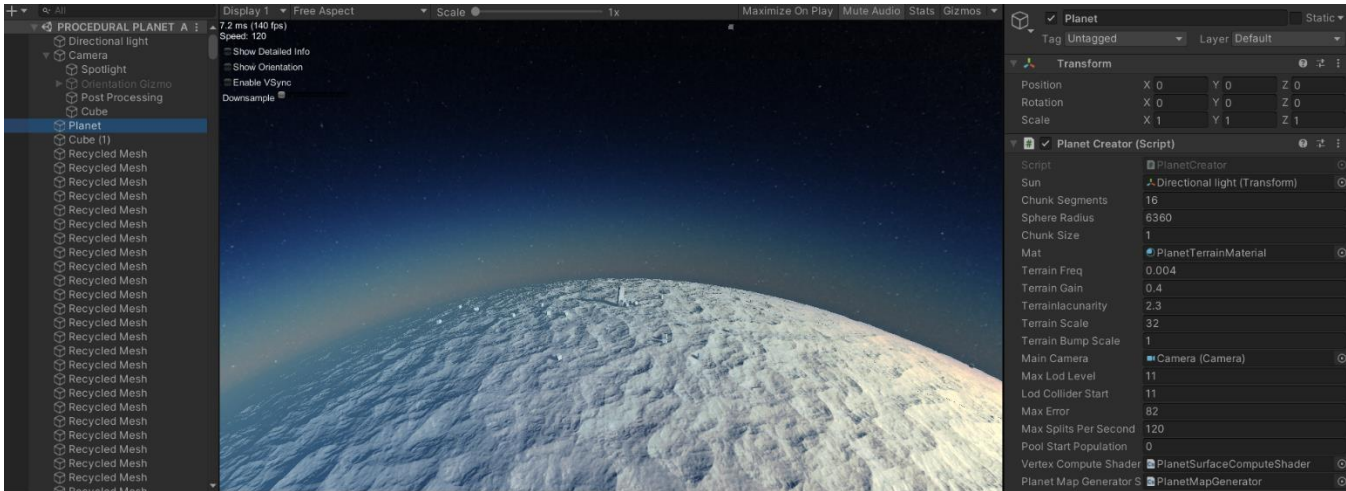


A system to export the procedurally created mesh that lives on the scene is available, using the “FBX Exporter” option shown in the following image.



Procedural Planet with LODs and physics interaction

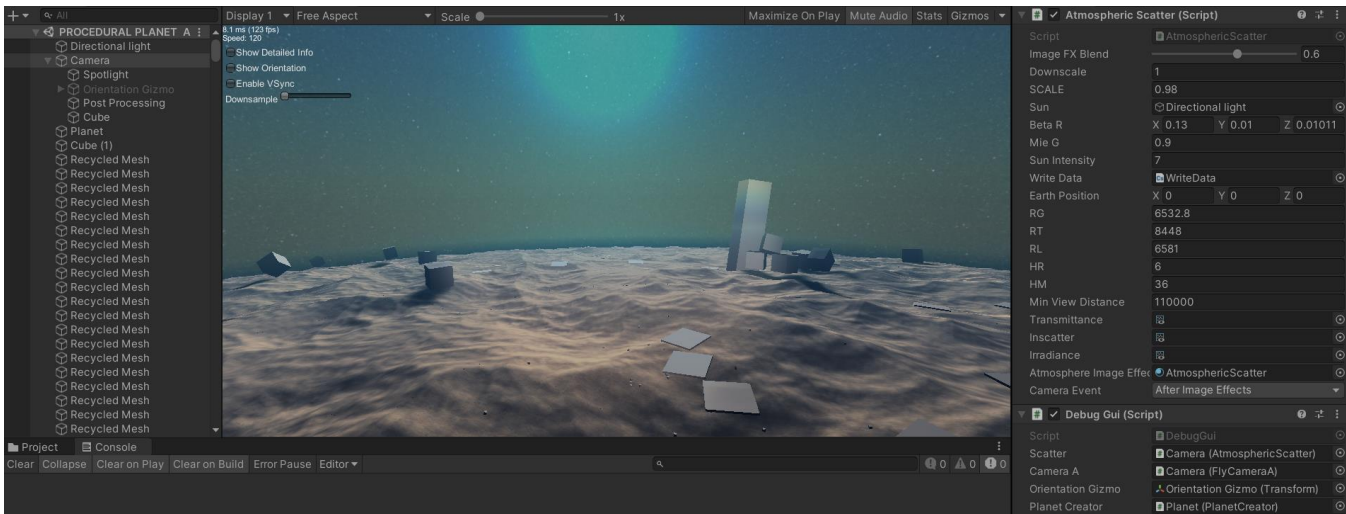
This module can be used to generate a planet on the fly, the various planet chunks are automatically updated and is possible to use collisions with the tiles for physics. A system to emulate planetary physics is also under evaluation for next versions. Use the “Sphere radius” to adjust the size of the planet.



Earth Atmosphere Rendering (Standard Pipeline, HDRP-URP under development)

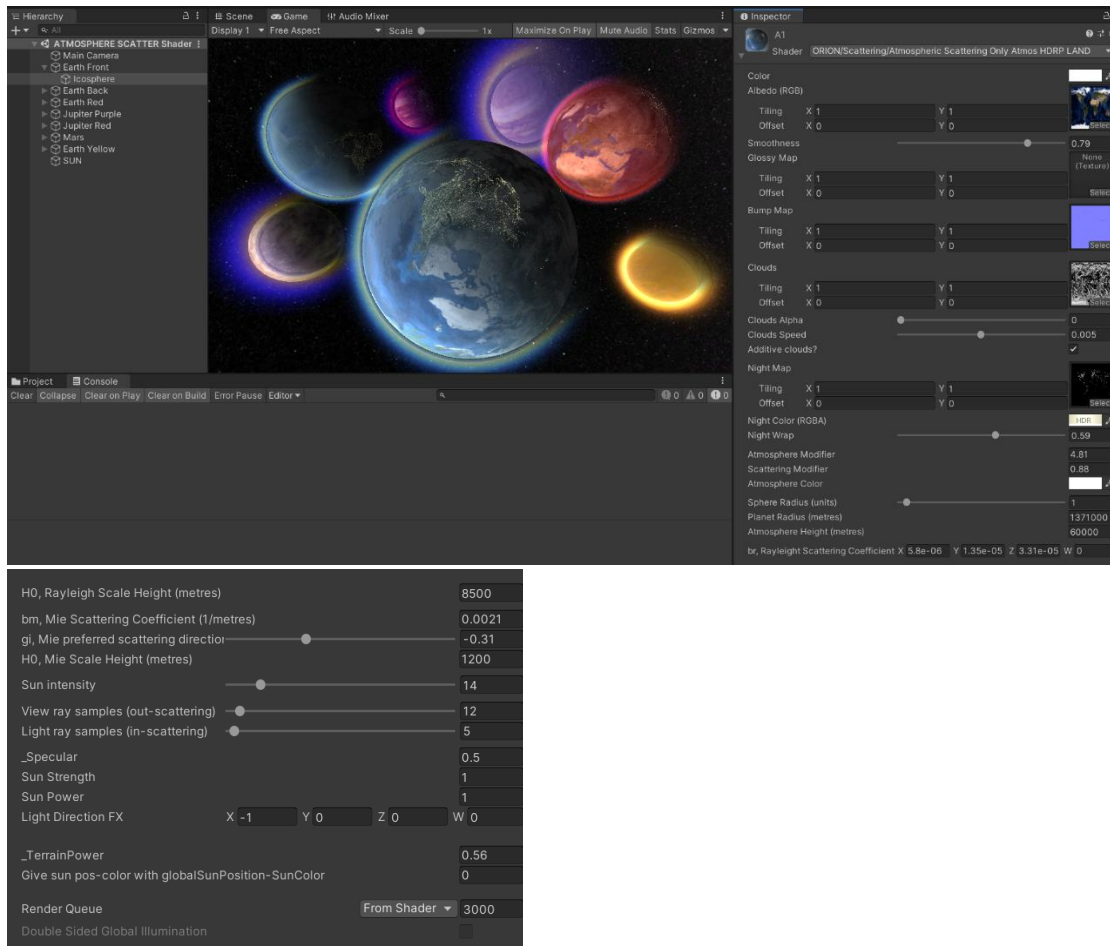


This module is for Standard Pipeline and can render a fully volumetric atmosphere around the planet, which supports fly through the volume and landing on the planet. Use the “Scale” variable to adjust the system based on the scale of the planet.

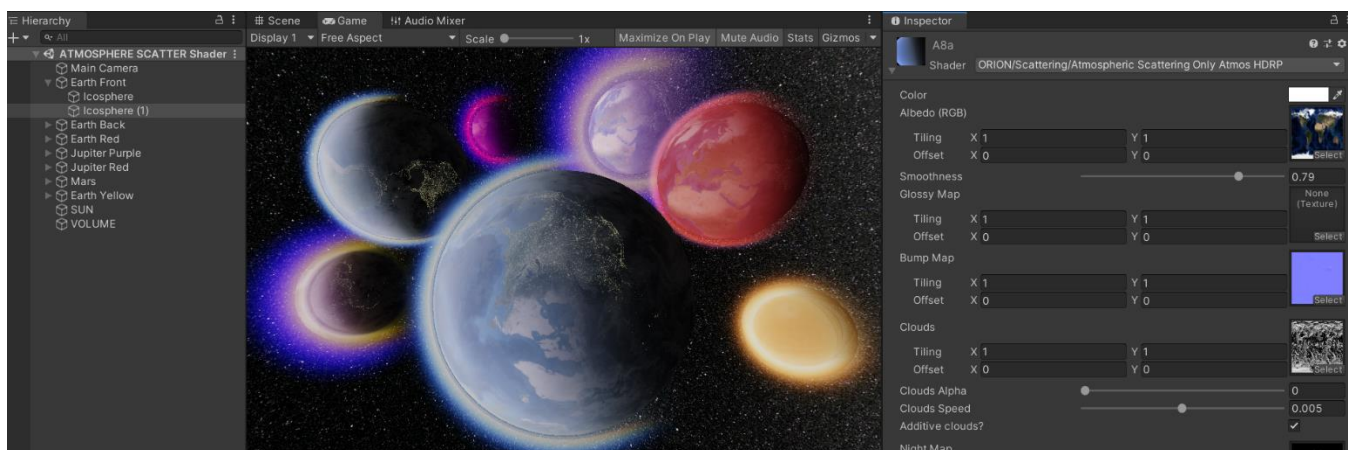


Atmospheric Scattering Shader for outside planet view

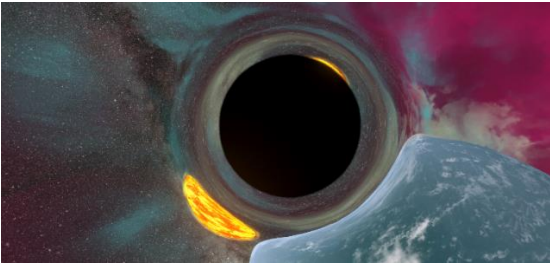
This module uses a volumetric atmosphere shader to emulate atmosphere using a sphere around the planet with the shader applied to that sphere. The “HDRP LAND” version can render both the atmosphere and planet (with night time in shadowed region) effect, while the “HDRP” version can be used to render the atmosphere alone around any scene planet. Note that for URP and HDRP the 2ond pass of the “HDRP LAND” shader version wont render, thus is needed to use the “HDRP LAND” version set at 2000 render order for the planet and the “HDRP” version applied to another sphere for the atmosphere.



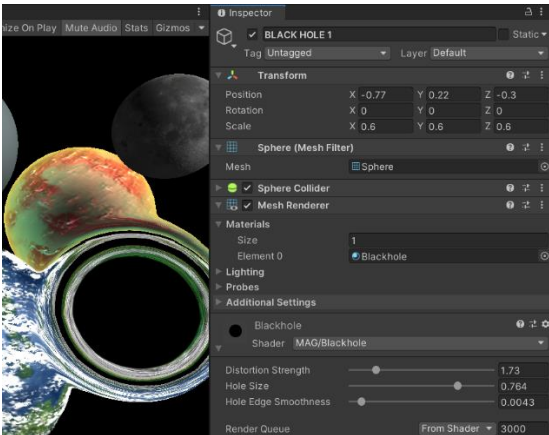
For URP and HDRP, only the first pass that renders the ground runs by default, thus must use a 2ond sphere for the atmosphere, using the atmosphere only shader as shown in the photo below.



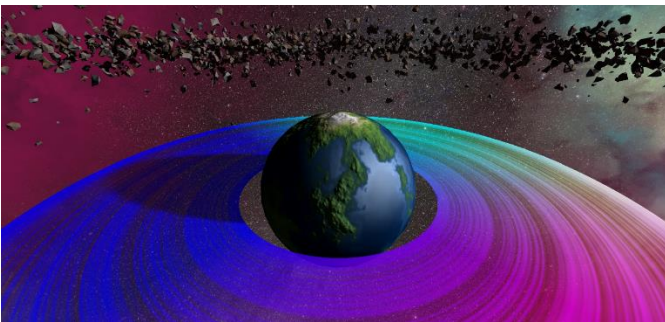
Black Hole



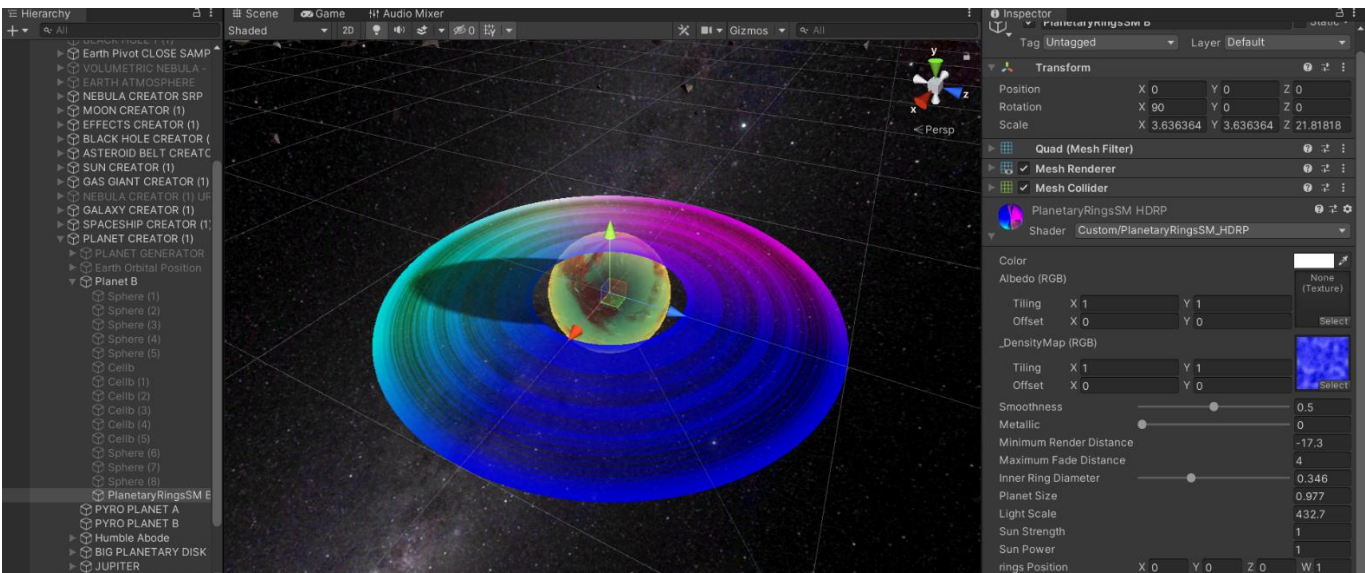
This module is a shader used on a quad to emulate a black hole. Version for all pipelines is available.



Planetary Disk



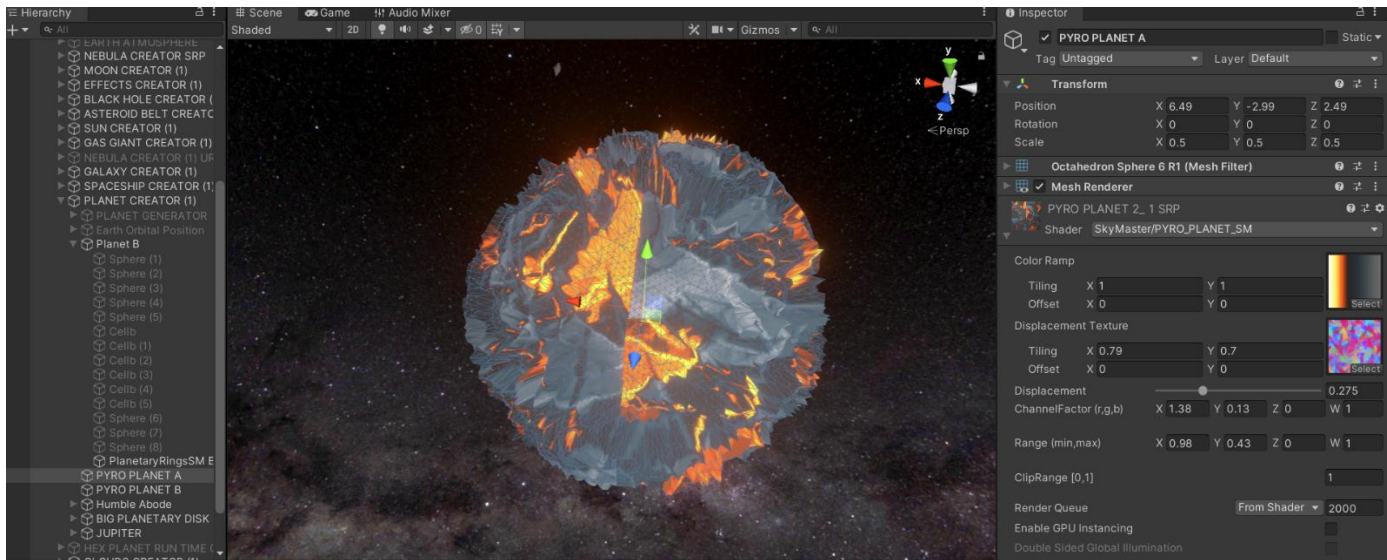
This module is a shader for planetary disks with shadow from the planet. Version for all pipelines is available.



Procedurally shaped planets



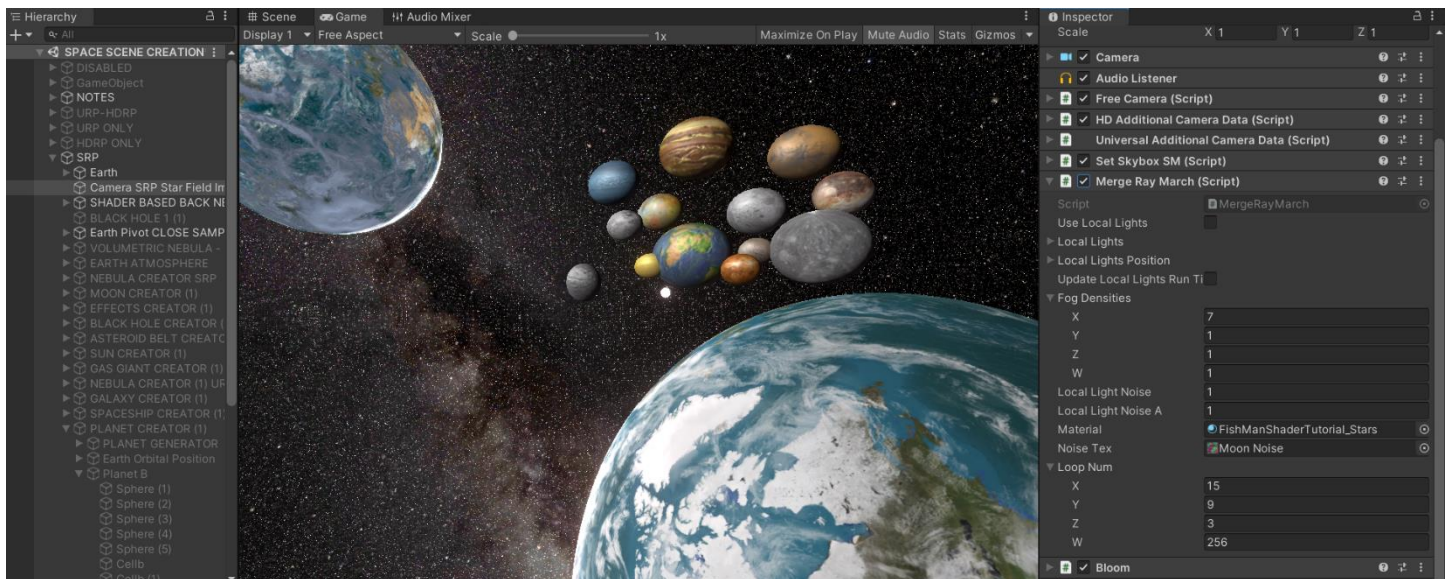
This modules consists of a shader that can manipulate the planet vertices at run time, for dynamic planet surface changes.



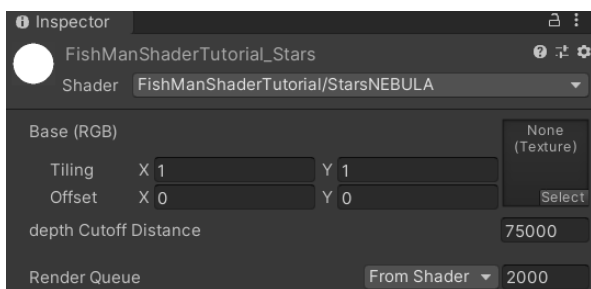
Stars (Image effect – Standard Pipeline)



This module can render a star field in the background as image effect and is available for the Standard Pipeline. The setup is by adding the “MergeRayMarch” script on the camera, using the proper material for stars in the Material slot, as shown in the image below.

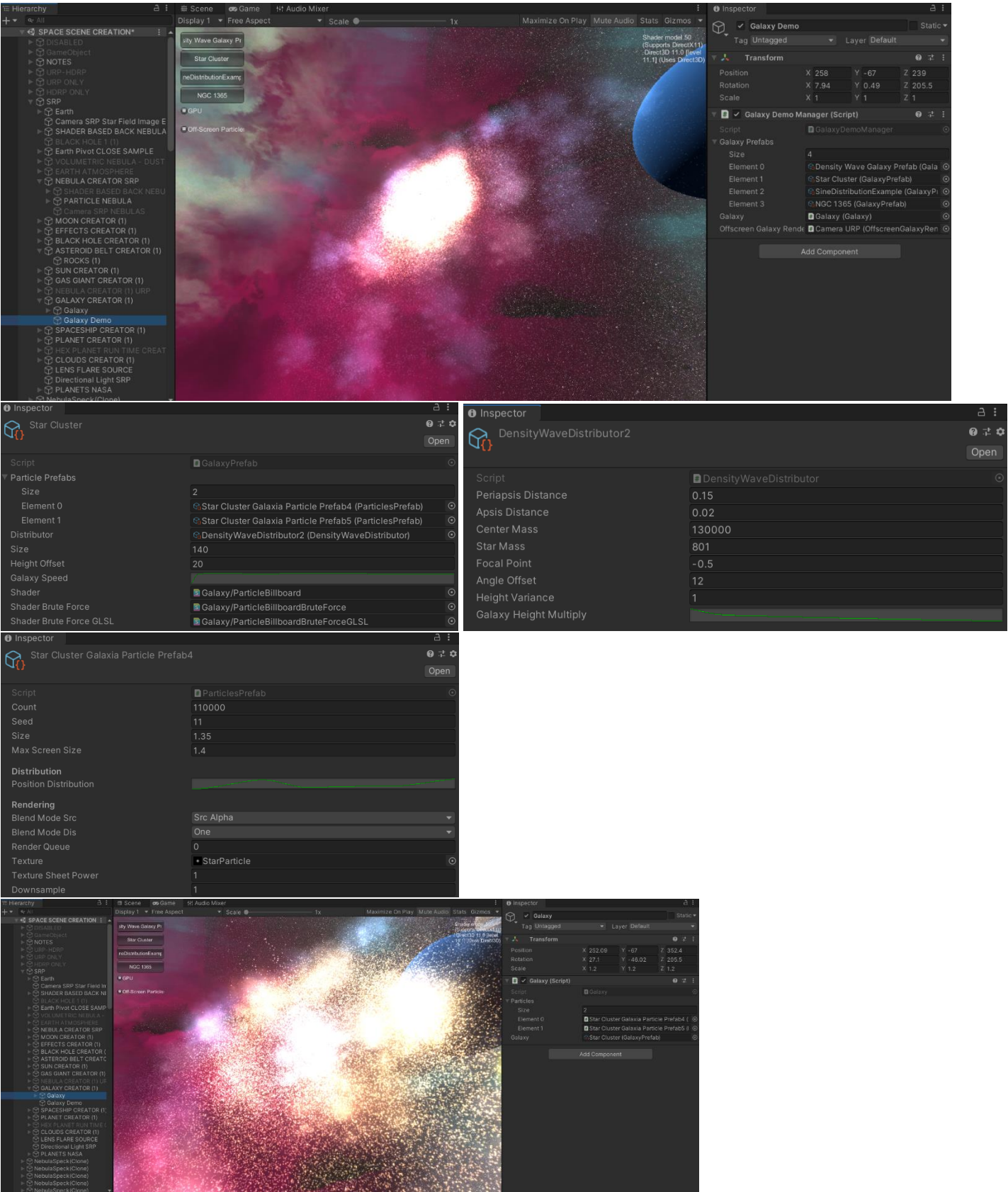


The material has a depth distance cutoff property that can be used to regulate how far the star field extends.



Galaxy Generator

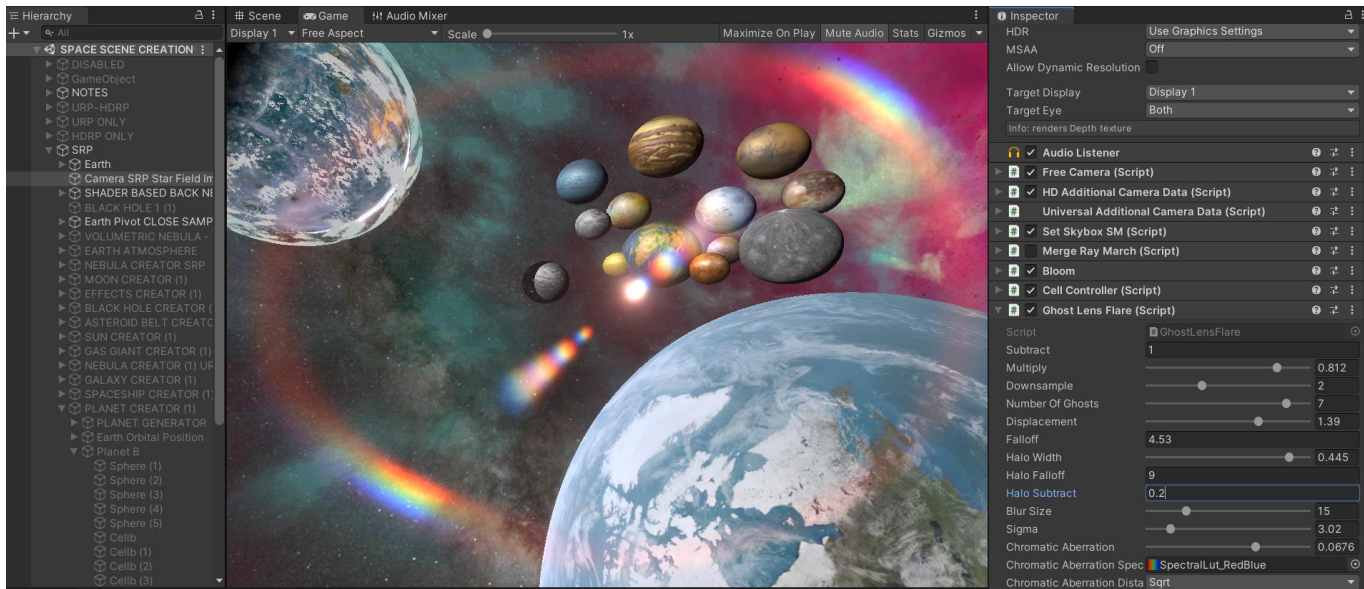
This module can be used to generate Galaxy formations, based on pre-defined settings and the galaxies can appear at run time.



Lens Flares (Image Effect - Standard Pipeline)

This module is an image effect based lens flare system, the system is setup using the “Ghost Lens Flare” script, which must be used on the main camera and is currently available for Standard Pipeline.

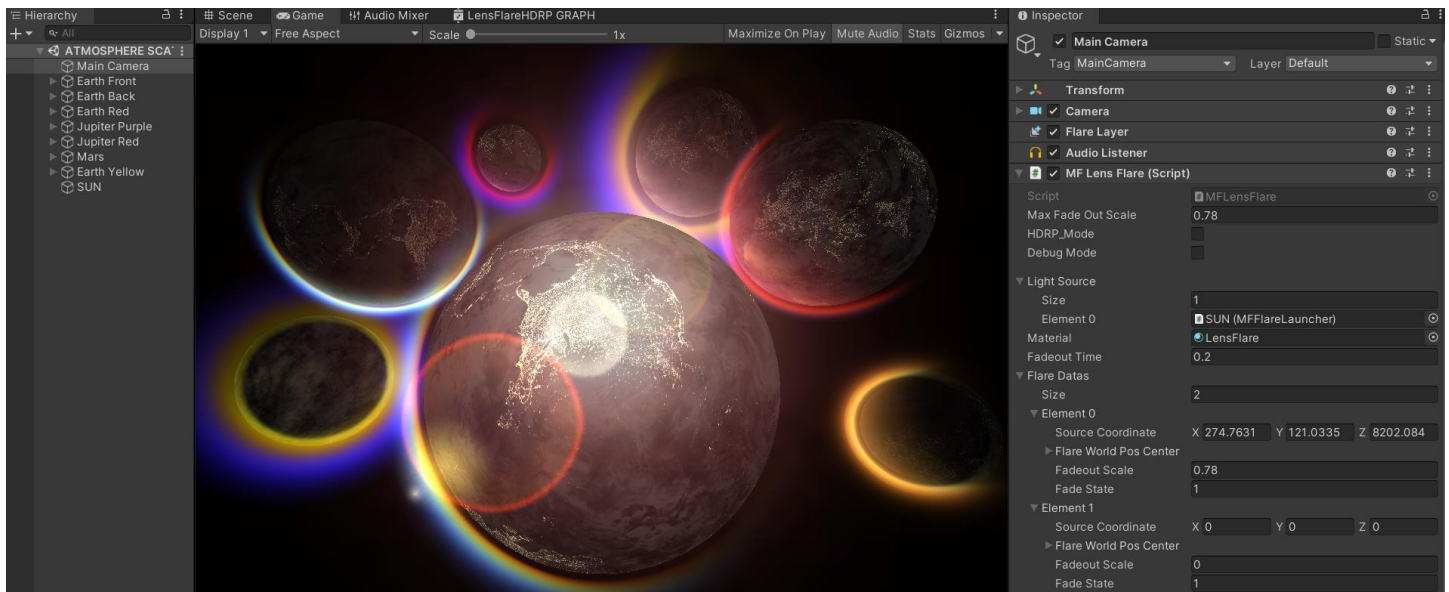
Note that for the system to function, HDR must be enabled on the camera settings.

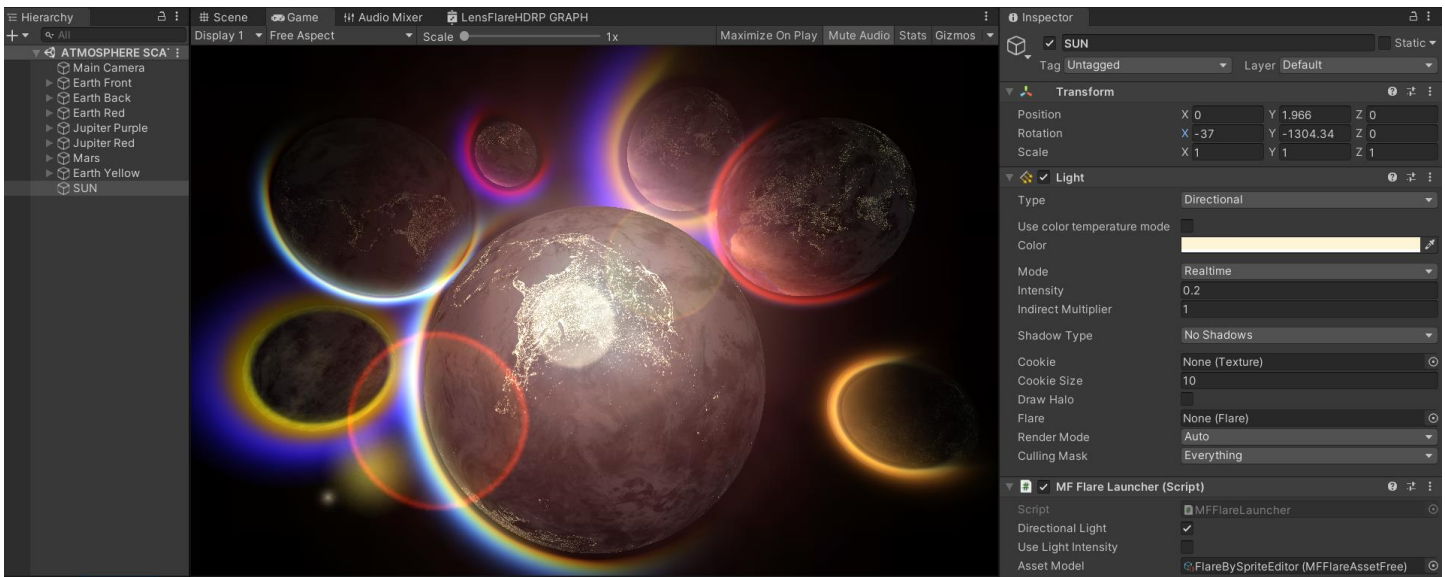


Lens Flares (Direct draw – All pipelines)

This module is a general lens flare system that does not rely on image effects and is applicable to all pipelines.

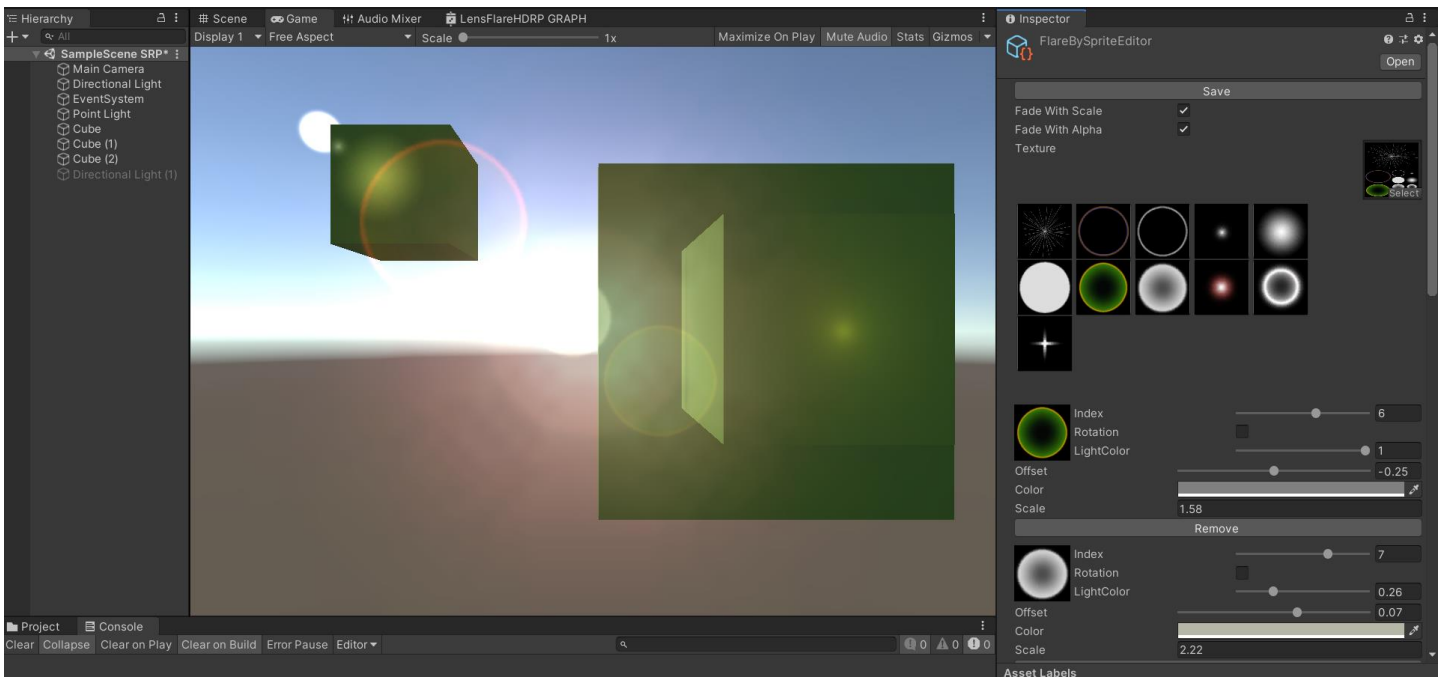
The setup consists of the “MFlareLauncher” script attached to the Main scene camera and the “” script attached to the scene Directional light.





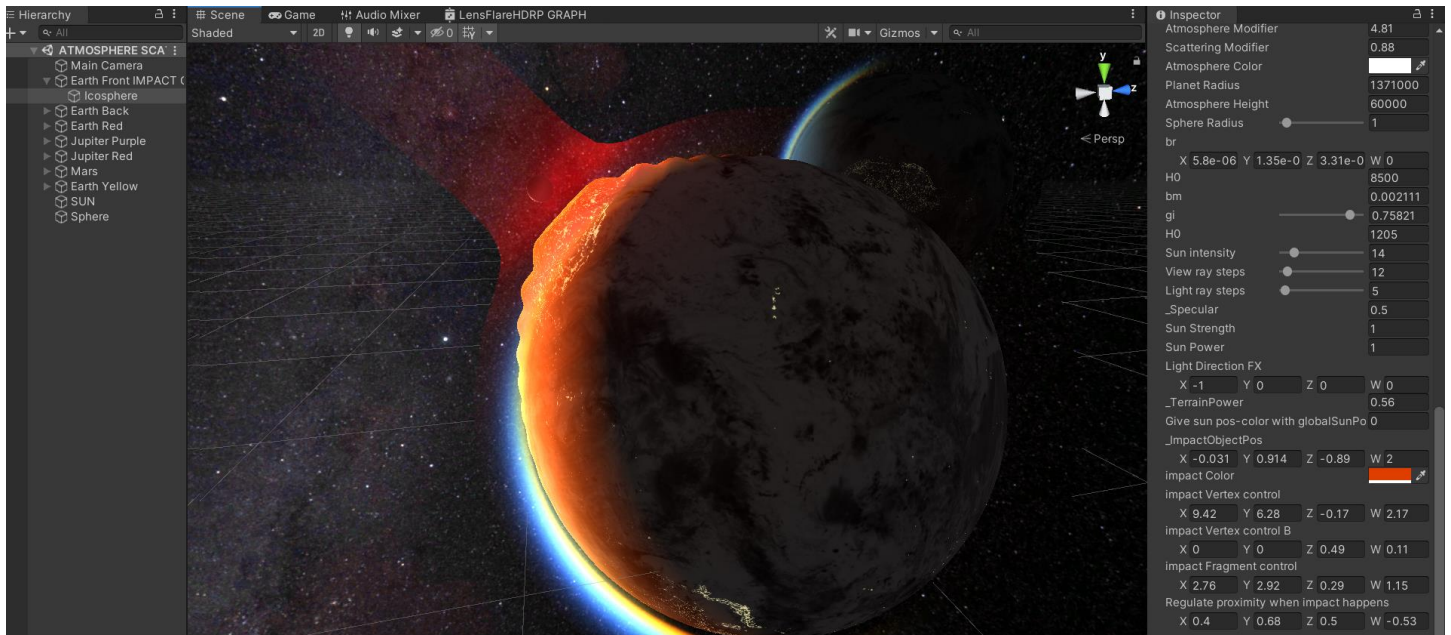
The lens flares asset models can be edited using the Unity 2D sprite editor, the “2D Sprite” package must be installed in the package manager for that purpose.

After the image is declared as sprite and the individual sprites are separated using the Unity sprite editor, then can assign the sprite in the “Asset Model” object “Texture” slot, by double clicking on it to open the lens flares editor and then choose the sprite and its distance to camera, color and size in the properties as shown in the following image.



Planet Impact

This module extends the atmosphere shader with an option to generate an impact like effect on the planet based on an impact object position. This is a first base of this system and will be further extended in next updates of ORION.



Aurora (This module is under development, manual will be updated after next asset update)

Hex Planet Generator (This module is under development, manual will be updated after next asset update)

Lightning (This module is under development, manual will be updated after next asset update)

